DEPARTMENT OF COMPUTER ENGINEERING · SENIOR DESIGN PROJECT 2026

A Systematic Comparison of Large and Small Language Models in *Standalone, Hybrid & Multi-Agent* Architectures

Doğukan Topçu · Abdulbaki Çeltik · Berkhan Kargılı · Meriç Kelemeden

Advisors: Dr. Hüseyin Ünlü · Prof. Dr. Onur Demirörs

İzmir Institute of Technology · İzmir, Türkiye · June 2026

1 Abstract / Summary

Measurement-first platform benchmarking **nine SLM–LLM architectures** across five reasoning benchmarks under a unified observability contract (accuracy · latency · cost · energy · carbon). Novel composite metric **EATS** (Efficiency-Accuracy Trade-off Score) jointly ranks architectures on quality and efficiency. **Routing** reaches 76.7% on MMLU with only 48.7% heavy-model calls; **speculative decoding** reaches 86.0%. Novel **Blackboard & Entropy Blackboard** resolve 72–93% of queries on SLMs alone at up to **97% accuracy** — highest in the study.

2 Background / Literature Review

LLMs deliver state-of-the-art reasoning but impose high latency, cost, and energy. **SLMs** are cheaper and data-sovereign, yet trail on broad reasoning tasks.

A PRISMA 2020-guided review of **54 collaborative SLM–LLM studies** reveals a critical gap: hybrid and multi-agent designs are rising, but **cost and energy are rarely first-class outcomes** alongside accuracy.

RQ1	**Performance.** How do collaborative SLM/LLM systems compare with monolithic LLMs on reasoning & classification?
RQ2	**Efficiency.** How do hybrid & multi-agent designs affect cost, latency, and energy?
RQ3	**Orchestration.** Which routing & division-of-labor mechanisms recur across architectures?
RQ4	**Domain fit.** Where do collaborative systems match LLM baselines — and where do they fail?

3 Measurement-First Platform

Every architecture consumes the same `Query` and returns the same `Response` carrying **accuracy, latency, tokens, cost, energy & CO_2** per query. The architecture layer is the *only* interchangeable component.

Experiment Config
architecture · benchmark · seed · models

↓

Control Plane
FastAPI · Next.js UI · runner · host-lock

↓

Architecture Layer
routing · speculative · oracle · ensemble · swarm

↓

Metrics Proxy + vLLM
latency · GPU power (NVML) · energy · CO_2 · cost

↓

Scorer → MLflow + Reports
per-item & aggregate metrics

TIER	HOST	VRAM	MODELS
SLM	GCP L4	24 GB	Gemma/Qwen/LLaMA 3–4B
Mid	RTX PRO 6000	96 GB	20–35B dense + MoE
Heavy	Nebius H200	141 GB	LLaMA-70B, gpt-oss-120B

4 EATS · Efficiency–Accuracy Trade-off Score

Accuracy α meets cost, latency & energy ratios normalised by the benchmark-specific standalone-LLM baseline.

$$P = 0.5\cdot\tilde{C} + 0.3\cdot\tilde{L} + 0.2\cdot\tilde{E}$$

$$\mathrm{EATS} = \frac{\alpha}{\alpha + P}$$

$\tilde{C} = C/C_0 \cdot \tilde{L} = L/L_0 \cdot \tilde{E} = E/E_0$

- Increasing in accuracy, decreasing in resource penalty; baseline reduces to **$\alpha/(\alpha+1)$** — a clean reference curve.
- Weights reflect deployment priorities: **cost > latency > energy**.

5 Energy, Carbon & Cost

A per-host metrics proxy times each request and samples GPU power via NVIDIA NVML — the boundary sits *at the inference host*: real power draw, not network overhead.

$$E = \frac{P_{GPU}\cdot\Delta s}{3.6\times10^{6}}\ \mathrm{kWh} \quad \cdot \quad CO_2 = kE \quad \cdot \quad C = C_{API} + r\cdot t$$

k = 380 gCO_2/kWh · GPU-tier rates \$0.75–\$6.50 / h

6 Benchmarks & Protocol

88 experiments, six architecture families, 100–520 samples per configuration, temperature 0.0, fixed seeds, shared prompts & parsers.

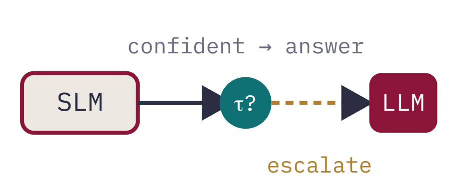
BENCHMARK	CAPABILITY DIMENSION
MMLU	world knowledge (57 subjects)
ARC-C	science reasoning
GSM8K	arithmetic reasoning (free-form)
HellaSwag	commonsense continuation
TruthfulQA	adversarial factual calibration

7 Architecture Taxonomy

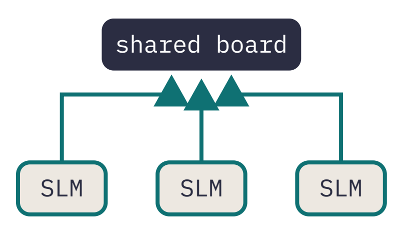
Ten architectures in three families. † novel designs from this project; ⋆ implemented but not yet benchmarked.

FAMILY	ARCHITECTURE	KEY MECHANISM
STAND	Single-Model	direct inference
STAND	MoE	sparse expert activation
HYBRID	**Routing**	confidence SLM→LLM escalation
HYBRID	**Speculative**	SLM drafts; LLM rewrites suffix
HYBRID	Debate (POA)⋆	proponent–opponent–arbiter
HYBRID	Active Oracle †	targeted LLM sub-queries
HYBRID	Blackboard †⋆	bid-based board; LLM sweeper
HYBRID	Entropy BB †⋆	uncertainty-penalised bidding
MULTI	Ensemble	confidence-weighted voting
MULTI	Pure Swarm †	SLM-only peers; no LLM online

ROUTING

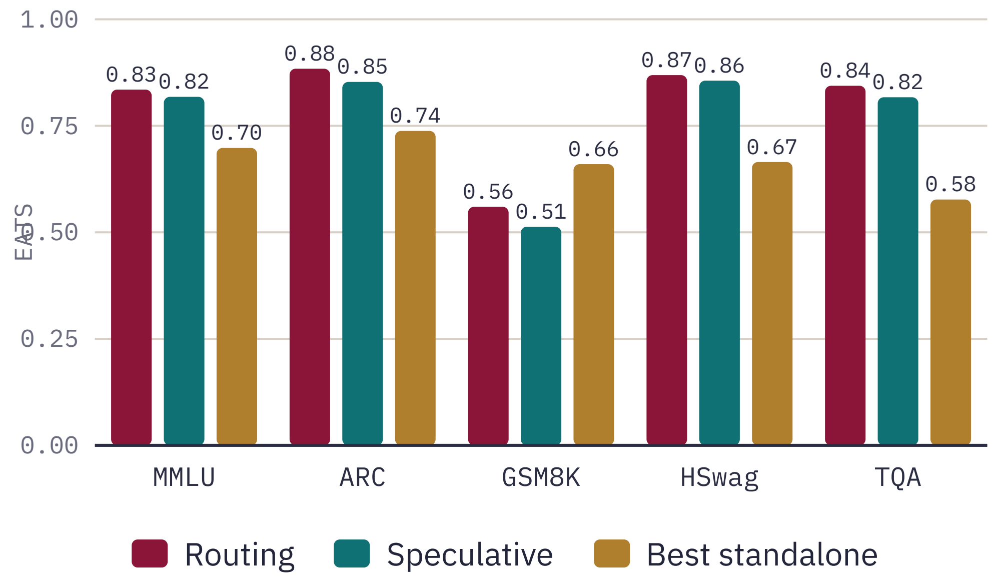


BLACKBOARD



8 EATS Frontier per Benchmark

Routing & speculative decoding (`gemma4-4b→31b`) take the top two EATS positions on **4 of 5 benchmarks**; on GSM8K lean MoE wins as routing escalates only 1% of queries.



9 Key Results

0.884 top EATS — **Routing on ARC** 91% acc · \$0.0005/query	**96%** **Pure Swarm** on ARC with **zero** LLM calls	**2–15×** cost & latency cut vs. standalone LLM

Routing across benchmarks — escalation is task-dependent, not fixed:

BENCHMARK	ACC %	EATS	LLM %	$/QUERY
MMLU	76.7	0.835	49	0.00074
ARC	91.0	**0.884**	75	0.00052
GSM8K	95.0	0.560	**1**	0.00255
HellaSwag	90.0	0.869	91	0.00061
TruthfulQA	85.0	0.844	81	0.00057

- **Speculative decoding:** 1–5 pts above routing (86% MMLU, 94% ARC) at EATS 0.82–0.86 — verifier rewrites only the divergent suffix.
- **Four-SLM ensemble:** highest raw GSM8K accuracy (96%), but 15–22s parallel latency caps EATS at 0.28–0.39.
- **Active Oracle** underperforms on multiple-choice: sub-queries add cost without recovering the answer chain.

10 Conclusions

RQ1	Routing & speculative **match best standalone accuracy** on ARC, HellaSwag & TruthfulQA; Pure Swarm exceeds every standalone on ARC.
RQ2	Hybrids cut per-query **cost 3–12× and latency 2–7×** at near-monolithic accuracy — gains are workload-dependent.
RQ3	EATS yields a **stable single-score ranking** of the quality–efficiency frontier across all families.
RQ4	Measured results corroborate **SLM-first escalation** & multi-SLM coordination trends from the 54-study review.

Future work

NEXT STEPS & SELECTED REFERENCES

Future work: benchmark Debate, Blackboard & Entropy Blackboard on the full suite · fix GSM8K numeric extraction · routing-threshold & SLM–LLM pair ablations · scale to ≥500 samples · edge deployment of the best-EATS architecture.

[1] Hao et al. **Hybrid SLM–LLM Inference**, 2024 · [2] Oh & Kim, **Uncertainty-Aware Hybrid Inference**, 2025 · [3] Shao et al. **Division-of-Thoughts**, 2025 · [4] Kwon et al. **vLLM: PagedAttention**, SOSP 2023 · [5] Companion SLR (54 studies, PRISMA 2020) · [6] MMLU, ARC, GSM8K, HellaSwag, TruthfulQA.

Code & experiments
github.com/DogukanTopcu